

Notebook

Day - 2.

Day-2.

Sample spaces and a classic puzzle.

1) Set Theory in Probability.

Probability measures the likelihood or chance of some event A happening.

A sample space Ω = set of all possible things^(outcomes) of a random experiment.

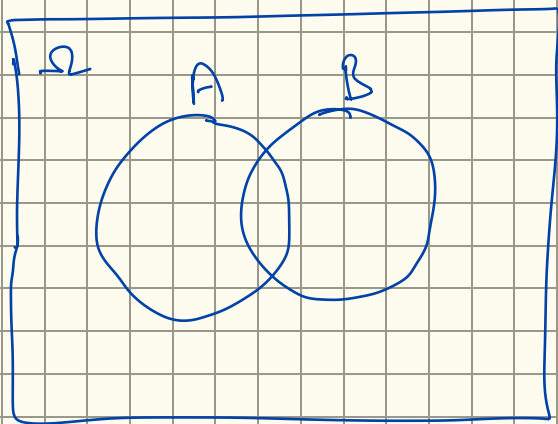
A specific element $\omega \in \Omega$ is one outcome (realization) of experiment.

Example: 3 coin flips.

$$\Omega = \{hhh, hht, hth, htt, thh, tht, tth, ttt\}$$

first flip is a head $A = \{hhh, hht, hth, htt\}$

Second flip is a tail $B = \{hth, htt, tth, ttt\}$



$A^c = \{\text{everything not in } A\}$

$A \cup B = \{\text{everything in } A \text{ or } B\}$

$$= \{hhh, hht, hth, htt, tth, ttt\}$$

$A \cap B = \{\text{everything in both } A \text{ and } B\}$

$$= \{hth, htt\}$$

[All are associative and commutative].

→ Probability (p) is a map from subsets of Ω to

$$\mathbb{R} \dots [0,1] \in \mathbb{R}. \quad [p: 2^\Omega \rightarrow [0,1]]$$

— we can have one biased & unbiased coin now.

properties:

1) $p(\Omega) = 1$

2) If $A \subset \Omega$ then $p(A) \geq 0$

3) If A and B are disjoint ($A \cap B = \emptyset$)

$$p(A \cup B) = p(A) + p(B)$$

follow ups:

$$p(A^c) = 1 - p(A)$$

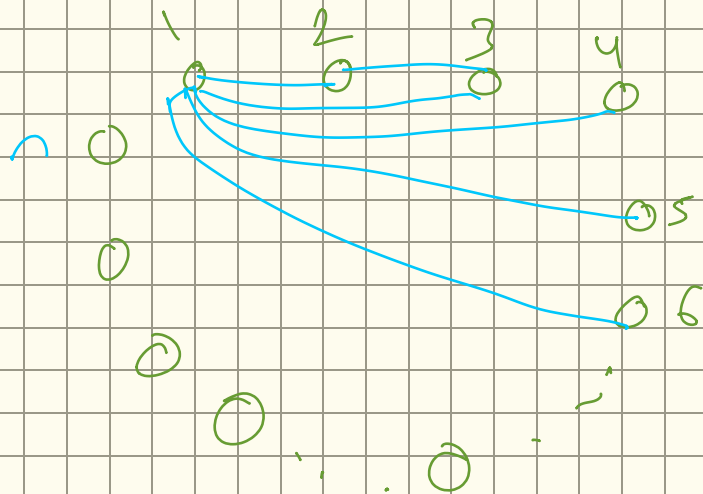
$$p(\emptyset) = 0$$

If $A \subseteq B$ then $p(A) \leq p(B)$

$$p(A \cup B) = p(A) + p(B) - p(A \cap B)$$

Birthday Problem.

If there are n people in a room, how large does n need to be for at least 50% chance of at least 2 people sharing a birthday.



Person 1 compares with $(n-1)$ people...

$$\begin{aligned} \text{Total} &= (n-1) + (n-2) + (n-3) + \dots + 3 + 2 + 1 \\ \text{Comparisons} &= \frac{n(n-1)}{2} \end{aligned}$$

If $n=20$, # of comparisons = 190

Let's Solve:

$$P(\geq 2 \text{ share}) = 1 - P(\text{no shares})$$

easier to calculate.

$$\begin{aligned} \text{For } P(\text{no shares}): & \frac{365}{365} \cdot \frac{364}{365} \cdot \frac{363}{365} \dots \frac{365-n+1}{365} \\ &= \frac{\# \text{ unique B-day}}{\# \text{ choices}} \rightarrow \frac{365!}{(365-n)!} \\ & \quad \quad \quad \hookrightarrow 365^n \end{aligned}$$

$$P(\text{no one shares}) = \frac{365!}{(365-n)! 365^n} \quad \leftarrow \text{closed form.}$$

$n = 23$

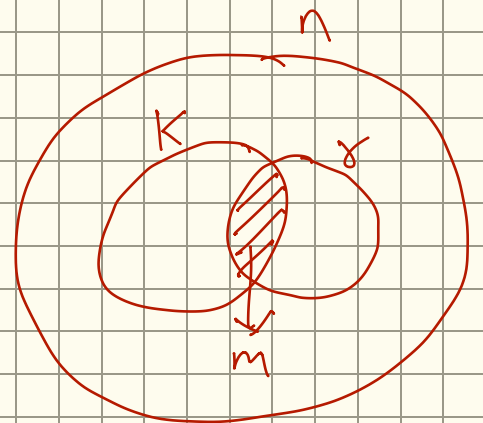
Quality Control (QC) and Non-Destructive Inspection (NDI)

n = total items in a lot.

K = # defective items.

r = sample size

m = # defectives in sample.



$P(r \text{ samples } m \text{ defective}) = ?$

$r = m \text{ defective} + (r-m) \text{ good.}$

$n = K \text{ defective} + (n-K) \text{ good.}$

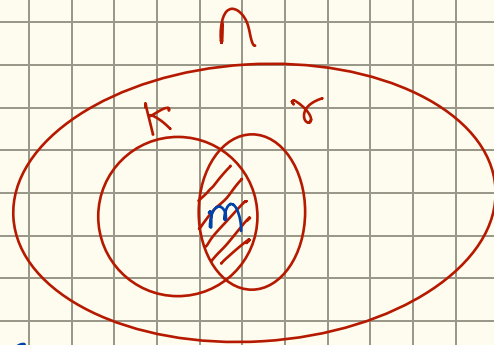
$$P(m \text{ defective in } r \text{ samples}) = \frac{\binom{n}{r} \binom{K}{m} \binom{n-K}{r-m}}{\binom{n}{r}}$$

r can be chosen from n

total # of r samples out of n
(order doesn't matter, w/o replacement).

ways of choosing defectives:

$$\binom{K}{m} \text{ total ways of sampling } m \text{ defectives from } K$$



$$\binom{n-K}{r-m} \text{ total ways of sampling } r-m \text{ non-defectives from } n-K. \text{ (it helps to describe the whole of the sample)}$$

product = Since they are independent.

$$p(m \text{ defective in } r \text{ samples}) = \frac{\binom{K}{m} \binom{n-K}{r-m}}{\binom{n}{r}}$$

Generalizing:

total:

$$N = G + B$$

(good) (bad)

sample:

$$n = g + b$$

(good) (bad)

$$p(b \text{ in } n \text{ samples}) = \frac{\binom{G}{g} \binom{B}{b}}{\binom{N}{n}}$$

→ Multinomial distribution.

→ To be more specific: hypergeometric distribution.

(Sampling without replacement from a fixed population split into two categories)

Statistics: getting K from N, r, n

using Monte carlo simulator:

For Birthday problem:

- number of trials (m)
- number of people in a room (n)

Steps:

Repeat ' m ' times

- 1) Assign ' n ' people random numbers between 1-365
- 2) Compare ' $n-1$ ', ' $n-2$ ', ... consecutively for all and keep a count.
- 3) If count > 0 then trial ' m ' $\rightarrow$ produced a match
else '0'

$\downarrow$
assign '1'
- 4) After ' m ' times $\rightarrow$ Take the average probability.